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Subcluster Operating Model in Managing Large Assets: Driving Efficiency Through People Empowerment

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Abstract

Asset M is the largest producing asset in South the Sultanate of Oman with over 30 producing fields, and faced operational complexities due to the high number of producing and injecting wells that need to be managed, the active drilling program, the diverse development types, and multiple production and injection stations. To address these challenges and enhance efficiency and performance, the asset transitioned from a functionally segmented, value stream-based model, where disciplines like the drilling team (New Oil), planning, and Well, Reservoir & Facility Management (WRFM) teams operated in silos, to an area-focused Subcluster Operating Model. This new model integrates all key activities within geographically defined zones, aiming to improve production performance, optimize costs, and empower teams through enhanced ownership, accountability, and cross-functional collaboration.

The Subcluster Operating Model organized the large unit (asset) into integrated area-based units called subclusters, each given full responsibility for its production system, including planning, execution, and performance monitoring. Resources such as rigs, hoists, and different units were allocated per subcluster to ensure operational flexibility. A centralized opportunity funnel was developed to mature development options and feed prioritized projects into execution systematically. Regular performance reviews, knowledge-sharing sessions, and cross-disciplinary collaborations were embedded to support the alignment and decision-making.

Moreover, the model significantly enhanced collaboration not only within subclusters but also among key stakeholders from other departments such as concept engineering, operations, and other supporting functions. This stakeholder engagement helps an integrated system approach to asset management, enabling smoother handovers and collective problem-solving. In addition to operational benefits, the Subcluster Model promotes talent development and leadership growth by empowering multidisciplinary teams to take ownership, build capabilities, and prepare future leaders.

To maintain strategic coherence and ensure that all subclusters pursue common goals, the overall operating model remains under the leadership and guidance of the Asset Manager and senior leadership team.

The implementation of the Subcluster Operating Model delivered substantial results across key performance areas. In terms of production growth, Asset M surpassed its production target ahead of

schedule, achieving the highest production levels since inception while reducing deferment by 15%. Additionally, the model accelerated the maturation of strategic future growth projects, ensuring the sustainable development of this large asset. Regarding cost leadership, subclusters operated with autonomy over allocated resources, such as hoists and standalone units, driving optimization and lean execution, which resulted in achieving the lowest Unit Operating Cost (UOC) in the past four years. Furthermore, the model promoted people empowerment and improved decision-making by embedding a culture of accountability, enabling teams to lead development planning and execution with enhanced technical and leadership capabilities. Decision making became faster and more effective due to improved visibility and alignment with long-term asset strategies.

The Subcluster Operating Model introduces a novel area-based operational structure within a complex & active asset environment. This approach decentralizes ownership and empowers multidisciplinary teams to manage full-cycle operations within defined geographies. It integrates planning, execution, and opportunity maturation within localized units, enhancing agility, resource optimization, and decision quality. This innovative model not only redefines operational efficiency but also sets a benchmark for a scalable, people-centric asset management system for large-scale assets.

Introduction

Managing large-scale oil and gas assets has always presented significant operational challenges, particularly when it comes to the allocation of critical resources, whether human capital, financial budgets, or physical infrastructure such as drilling rigs and hoists. Asset M, the largest producing asset in the Southern region, serves as a prime example of these complexities. With over 30 producing reservoirs, a diverse range of development types, a large number of existing well stocks, active drilling programs, and multiple production and injection surface facilities, the asset faces continuous operational and logistical difficulties. The volume of simultaneous activities from drilling and maintenance to optimization and expansion makes resource allocation an ongoing challenge, requiring careful prioritization to balance value creation, cost efficiency, and operational effectiveness.

These complexities often lead to prolonged decision-making cycles, as stakeholders must weigh competing demands and work on priorities. The resulting delays not only hinder immediate performance but also contribute to long-term value erosion. Furthermore, misalignment in resource distribution can have falling effects, including underutilized assets, production inefficiencies, and inflated operational expenditures. Without a structured approach, even high-potential assets may underperform, failing to meet production targets or maximize returns.

Recognizing these challenges, Asset M pioneered the Subcluster Operating Model a tailored framework designed to streamline decisionmaking, enhance accountability, and optimize resource deployment. By decentralizing operational oversight while maintaining strategic alignment, this model has proven effective in addressing the inherent difficulties of managing such a vast and multifaceted asset. This paper will examine the development, implementation, and outcomes of the Subcluster Model, highlighting its role in transforming Asset M's operational efficiency and long-term performance.

Problem Statement

Initially, Asset M was structured around a conventional value stream-based operating model, consisting of three primary value streams (as shown in [Figure 1](#)). The first stream was Well and Reservoir Management (WRM), which was responsible for managing reservoir production and injection, handling deferment, planning and executing well interventions, optimizing production performance, and overseeing both CAPEX and OPEX expenditures. The second stream, Development Planning, focused on managing reserves, generating production forecasts, and leading hydrocarbon maturation activities to ensure a steady pipeline of future development opportunities. The third stream, New Wells Delivery, was tasked with

planning and executing new drilling activities that were central to unlocking growth and sustaining long-term production targets.

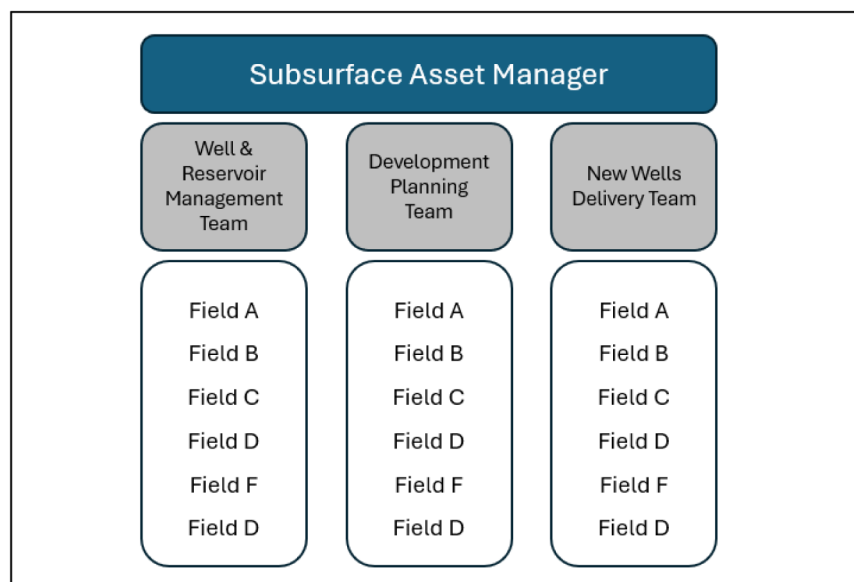


Figure 1—Diagram describing the initial conventional value-stream approach adopted by Asset-M

While the value stream-based model provided clear functional boundaries and accountability, it faced several structural and operational challenges. One major limitation was the lack of integration between the streams, which led to organizational silos and limited collaboration. This segmentation often resulted in centralized, slow decision-making, as coordination required multiple layers of alignment and approvals. The model also fostered a top-down environment with limited empowerment of frontline teams, contributing to low motivation and reduced people efficiency. The absence of ownership and autonomy at the working level hindered innovation, delayed responses to operational issues, and constrained the asset's ability to adapt swiftly to emerging challenges. In addition, functional teams often focused on optimizing their own KPIs, which sometimes led to misaligned objectives and suboptimal outcomes at the asset level. Opportunity maturation was frequently slowed due to handovers between streams, with unclear ownership during transitions. The rigid structure limited responsiveness to localized subsurface complexity and offered few avenues for cross-functional learning or innovation at the frontline. This also created challenges in developing well-rounded talent with integrated operational insight, as most individuals remained confined within their functional silos. Given the increasing pace and complexity of activities within Asset M, it became evident that a more dynamic and responsive operating model was needed. The asset required an improved structure that would accelerate decision-making, enhance integration, and boost operational efficiency. Moreover, there was a pressing need to improve people's performance by introducing a culture of healthy competition, recognizing individual contributions, and empowering teams to take ownership of planning and execution. Such a transformation was not only critical for immediate performance gains but also essential for building future leaders and sustaining long-term success in a challenging and fast-paced operating environment.

Proposed Subcluster Model

Asset M introduced a new organizational structure known as the Subcluster Operating Model. This model marked a significant shift from the traditional, functionally segmented value stream-based approach to a more integrated and geographically focused structure. Instead of separating responsibilities across distinct value streams, the asset was restructured into area-based units called subclusters (Figure-2), each responsible

for the end-to-end management of activities within a defined geographic or reservoir area. The subcluster approach aimed to create a more agile, collaborative, and empowered operating environment, better suited to handle the high volume and diversity of activities within Asset M.

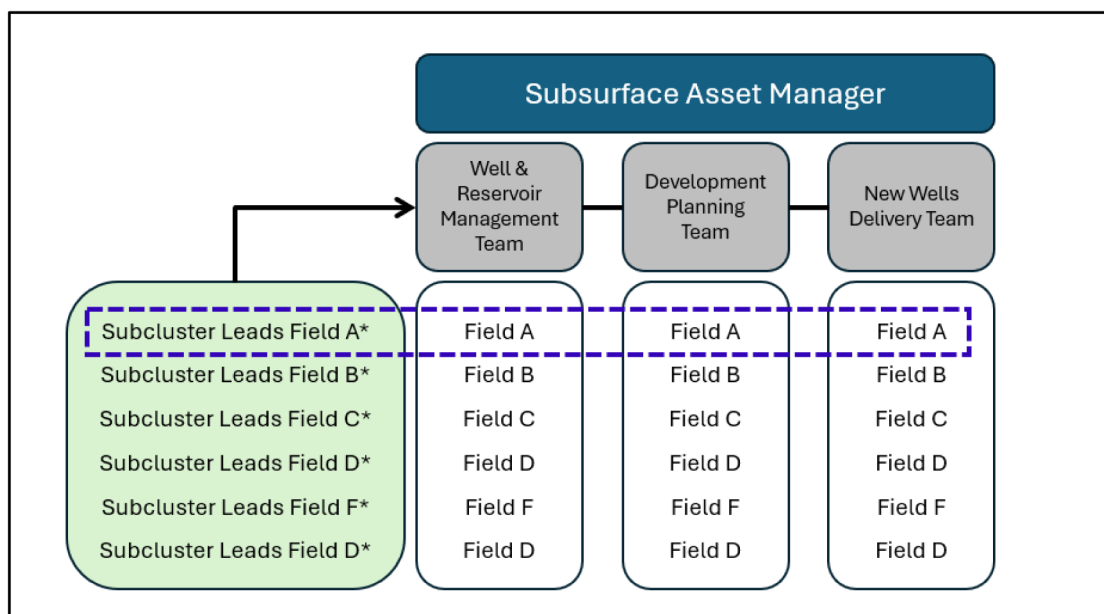


Figure 2—Diagram describing the area-based subcultures operating model adopted by Asset-M . *Subcluster lead are senior team members from each discipline.

Each subcluster team is a multidisciplinary unit, incorporating representatives from all core functions: Well and Reservoir Management, Development Planning, and New Wells Delivery under the guidance of senior subcluster lead from each discipline. These subcluster leads are tasked not only with technical role but also with driving integration and alignment across functions. This structure enables a more responsive and cohesive decision-making process, as team members work side by side with a shared understanding of local challenges, priorities, and opportunities. By breaking down silos and decentralizing control, the model empowers subcluster teams to take full ownership of their business area, enhancing both accountability and engagement.

Under this model, subclusters are fully responsible for managing the business and delivering results within their allocated area, including production, injection, deferment management, well interventions, hydrocarbon maturation, and drilling activities. This autonomy fosters a culture of proactive leadership, stronger collaboration, and quicker execution, enabling teams to respond rapidly to operational issues and unlock more value from their activities.

The Subcluster Operating Model has also significantly enhanced cross-functional alignment and stakeholder engagement across the asset. By organizing teams around geographic and reservoir-based areas with clear ownership, the model fosters tighter collaboration with key stakeholders beyond the core WRFM and development teams. For instance, Well Engineers are engaged early and consistently throughout the well planning, design, and execution stages, ensuring that subsurface insights are fully integrated into well delivery, and that lessons learned are rapidly captured and applied. Similarly, Surface and Concept Engineers, along with Operations teams, are closely embedded in subcluster discussions related to facility setup, debottlenecking, hook-up readiness, and production system optimization. This operational proximity ensures that facility designs are optimized, implementation challenges are addressed early, and handovers are smooth and efficient. The result is a more coherent, end-to-end planning and execution cycle that breaks down functional silos, improves schedule fidelity, and ensures that both subsurface and surface

constraints are addressed holistically. This level of integration not only drives better technical outcomes but also strengthens ownership, alignment, and mutual accountability among all contributing teams.

The transition to the Subcluster Operating Model was supported by a structured change plan that helped ensure a smooth and practical shift without disrupting operations. The Asset Manager and leadership team led the change by clearly explaining the purpose of the model, aligning the teams on the expected outcomes, and closely supporting them during the early stages. Instead of formal training, on-the-job coaching, hands-on support, and regular alignment sessions were used to help teams gradually build the confidence and capabilities needed to operate in the new setup. Regular check-ins and feedback loops were used throughout to capture learnings, address challenges early, and adjust the approach as needed. This practical and people-focused approach helped embed the model successfully and build ownership within the teams.

To support the effective rollout and sustainability of the Subcluster Operating Model, all performance monitoring tools and dashboards were redesigned to reflect the new structure. These tools now provide real-time visibility of key performance indicators such as production rates, injection performance, deferment trends, and the status of well workovers and new development projects at the subcluster level. This transparency enables data-driven decision-making and gives teams the ability to track their performance and identify improvement opportunities independently. As a result, subclusters are not only better positioned to manage day-to-day operations but are also empowered to drive continuous improvement and long-term value creation across the asset.

Despite the subclusters operating with a high degree of autonomy, reporting lines remain intact through the subcluster leads to the respective team leads and the Asset Manager. To maintain alignment, ensure consistent performance management, and foster a culture of shared accountability, a structured cadence of both formal and informal reviews has been established. This includes a fixed calendar of weekly performance review sessions involving all subclusters, where short-term results, operational priorities, and emerging issues are discussed in a collaborative forum. In addition, dedicated deep-dive sessions are conducted regularly to assess each subcluster's performance end-to-end, covering production delivery, deferment trends, well activity execution, and hydrocarbon maturation progress. These reviews not only facilitate transparency and timely intervention but also create a valuable platform to surface technical and nontechnical challenges, share lessons learned, and align on cross-functional support needs.

Results and Discussion

By adopting the Subcluster Operating Model, Asset M has consistently demonstrated that agile, integrated, and ring-fenced teams can deliver exceptional performance year after year. This transformation has proven that when personnel are empowered, aligned, and held accountable within a well-defined structure, they aim high and often exceed expectations. The impact of this model is clearly reflected in several key performance areas, beginning with production growth. Under this model, production reached a record-breaking milestone ahead of schedule, with deferment reduced by 15% a strong testament to the model's effectiveness (Figure.3).

Subcluster teams take full ownership of their production systems, ensuring seamless operations, rapid issue resolution, and a continuous drive for improvement. This deep sense of accountability has been instrumental in maintaining high efficiency and quality across all production processes. Additionally, the model has enabled the successful delivery of a large number of wells, with strict adherence to the well delivery process and excellent compliance, reinforcing both operational discipline and output quality. Beyond immediate performance, the model has also accelerated the maturation of future growth opportunities, demonstrated by the advancement of critical development projects and ongoing appraisal efforts that lay the foundation for sustained production and long-term asset growth.

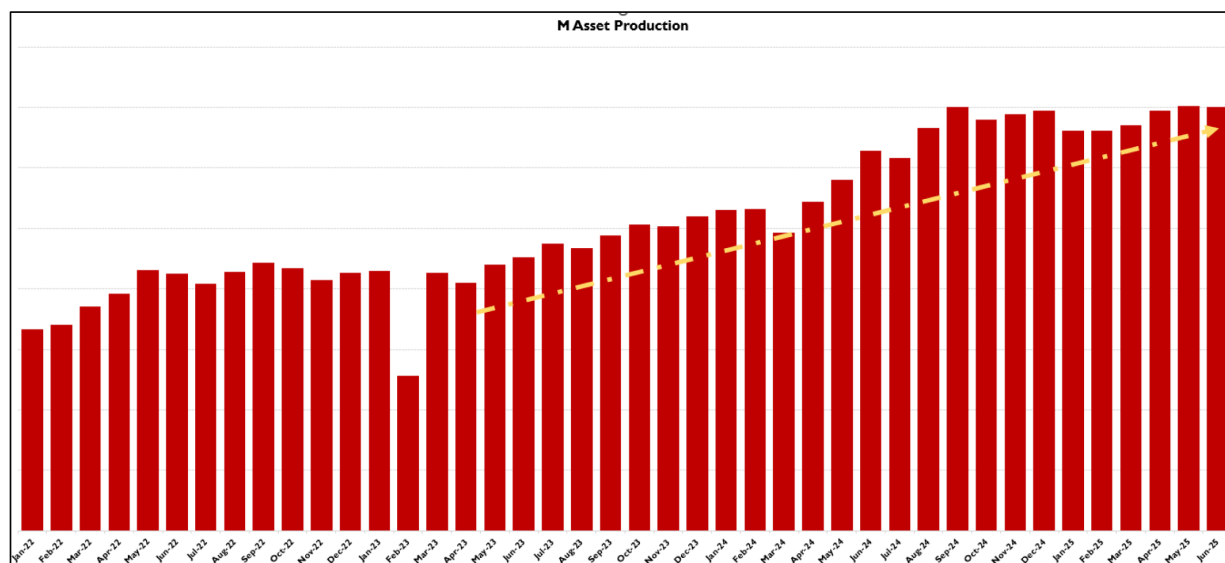


Figure 3—Production Trend of Cluster M

In terms of cost leadership, the model has empowered subclusters to manage their activities more efficiently and mature opportunities through a structured opportunity funnel. This funnel system ensures that only high-value opportunities are progressed and resourced appropriately, leading to the best-recorded Unit Operating Cost (UOC) in the past four years (Figure.4).

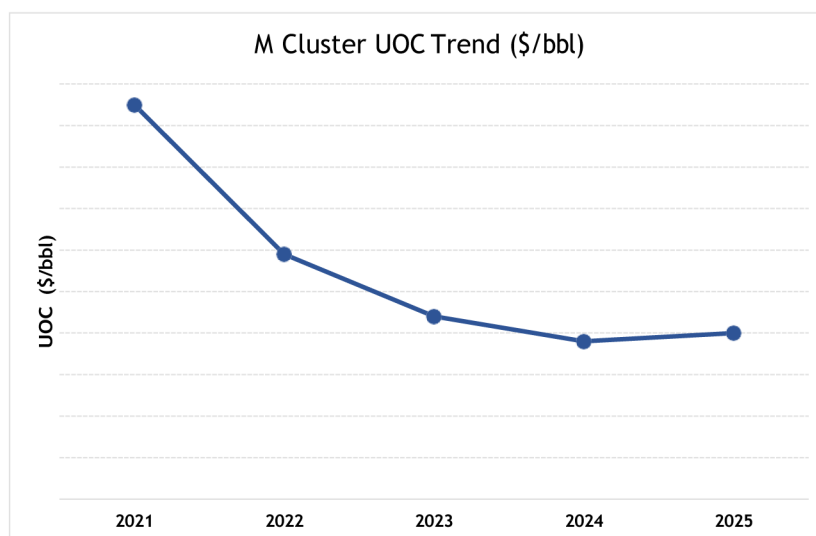


Figure 4—UOC Trend for Asset M over the past years

This achievement reflects not only disciplined execution but also strategic resource allocation aligned with the asset's broader objectives.

Equally important is the model's focus on people empowerment and effective decision-making. A culture of collaboration, innovation, and knowledge-sharing has been deeply embedded within subclusters, significantly enhancing both technical depth and leadership strength. Teams are not only empowered to lead day-to-day execution but also contribute meaningfully to long-term planning. Each subcluster has been instrumental in shaping and aligning its five-year development plans with the overall asset strategy, ensuring coherence across short- and long-term goals. The ability to manage both personnel and physical resources such as rigs and hoists at the subcluster level further reinforces agility and operational control.

As the Subcluster Operating Model evolves, it's important to keep building the skills and leadership needed to support it in the long run. While teams are working well within their areas, giving people the chance to rotate between subclusters or take on short-term roles in other areas can help them gain broader experience and learn new ways of working. This also helps share good practices across the asset and keeps teams flexible. At the same time, it is the responsibility of the team leads and the Asset Manager to ensure strong future leaders are developed. Subcluster leads need more than just technical skills—they also need to guide their teams, solve problems, and think about the bigger picture. Supporting their growth will help keep the model effective and ready for future challenges.

Overall, the Subcluster Operating Model has emerged as a powerful enabler of performance, resilience, and sustainable growth for Asset M.

Conclusion

In conclusion, the transition to the Subcluster Operating Model has fundamentally transformed Asset M's operational framework, effectively addressing the multifaceted complexities inherent in managing one of the largest and most diverse oil and gas assets in the region. By moving away from the traditional, siloed value stream-based approach to a more integrated, area-focused structure, this model has successfully decentralized ownership, empowered multidisciplinary teams, and fostered a culture of accountability within geographically defined subclusters. This structural shift has led to remarkable improvements across key performance metrics, including achieving record-breaking production levels ahead of schedule, significantly reducing production deferment by 15%, and attaining the lowest Unit Operating Cost (UOC) seen in recent years. These achievements not only reflect enhanced operational efficiency but also demonstrate the model's ability to drive sustainable growth through effective resource allocation and optimized execution.

Furthermore, the Subcluster Operating Model has introduced a collaborative and innovative culture that deeply embeds technical excellence, leadership development, and cross-functional knowledge sharing at all organizational levels. By empowering teams to take ownership of both daily operations and long-term development planning, the asset has improved the speed and quality of decisionmaking, aligning operational activities closely with strategic objectives. The ability to manage critical resources such as personnel, rigs, and hoists within each subcluster has further increased agility and responsiveness, enabling the asset to adapt rapidly to changing market conditions and operational challenges. This integrated approach has not only enhanced performance resilience but also positioned Asset M as a benchmark for excellence in large-scale asset management.

Ultimately, the Subcluster Operating Model serves as a powerful, scalable framework that can be replicated by other complex assets seeking to overcome similar operational challenges. By fostering a people-centric, accountable, and strategically aligned operating environment, it offers a roadmap for achieving superior efficiency, cost leadership, and sustainable growth in today's dynamic and competitive energy landscape. This model exemplifies how innovative organizational design can unlock value and drive long-term success in large oil and gas operations, setting a new standard for industry best practices.

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